

System Driven Hardware Design

Universal Asynchronous Receiver Transmitter (UART)

International Master of Science

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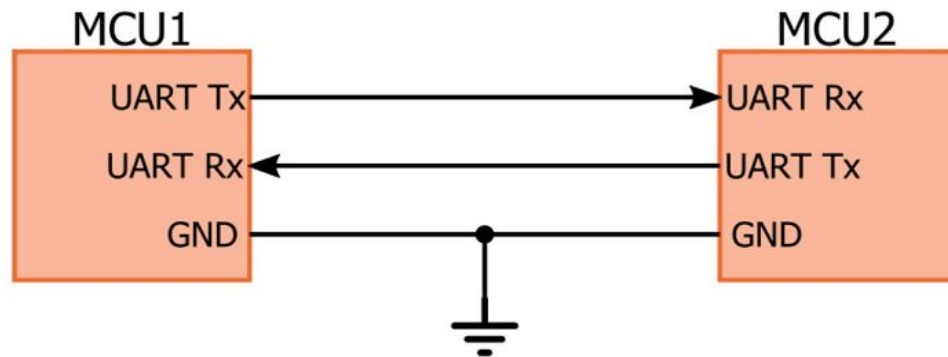
UART helps to exchange data to the Personal Computer

- Helps Algorithm testing
- Helps debugging
- Performance testing

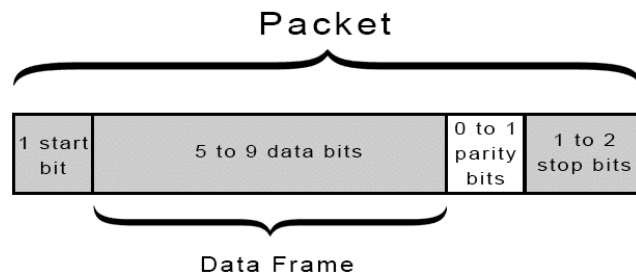
Agenda

- o UART
- o Exercise

Hardware Wiring



One Transmission Packet



[Hardware Details: All About Circuits](#)

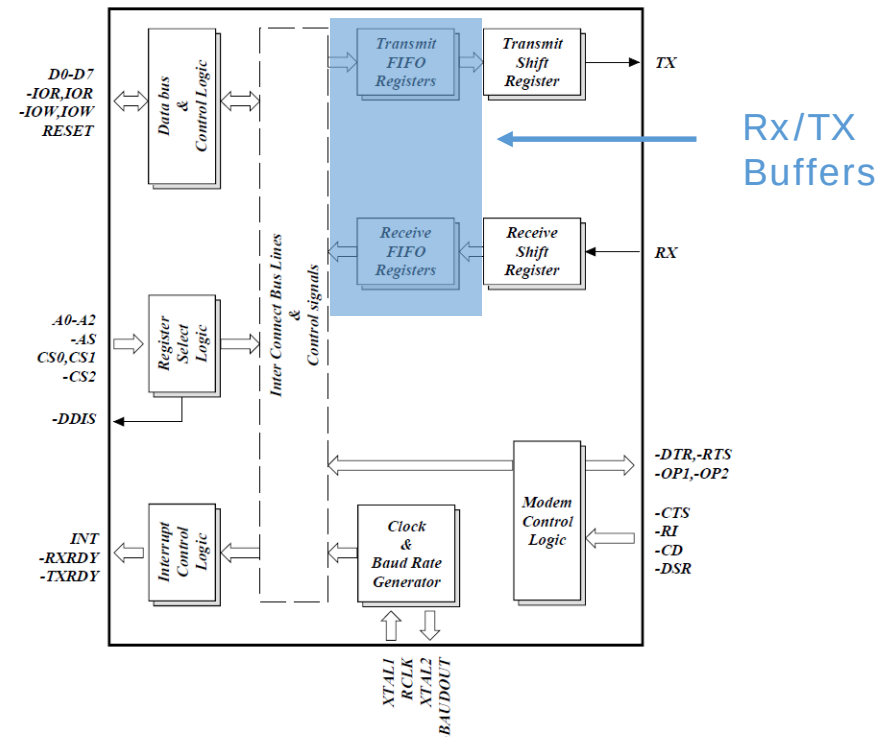
Note

- RX – TX is crossed

Settings at Transmitter and Receiver

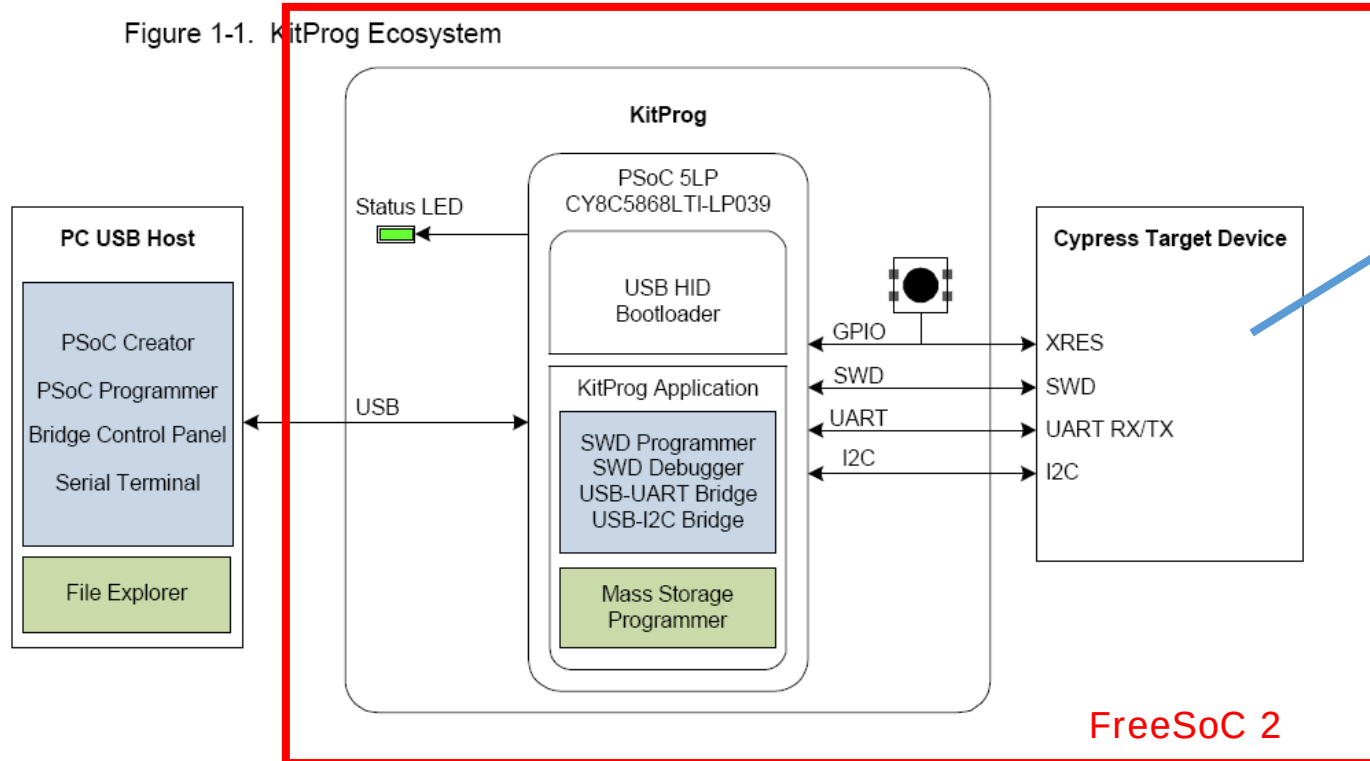
- Asynchronous transmission, hence Baudrate has to be set, since the clock is recovered at the receiver
- Packet Composition must be known

Internal UART block diagram (courtesy of the Exar ST16C550 datasheet)

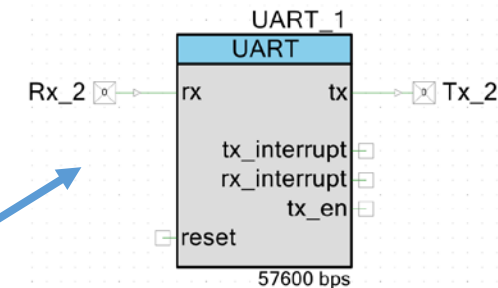


FreeSoC 2: USB – UART Bridge

Figure 1-1. KitProg Ecosystem



Top Design View



Hints for
Your
Project

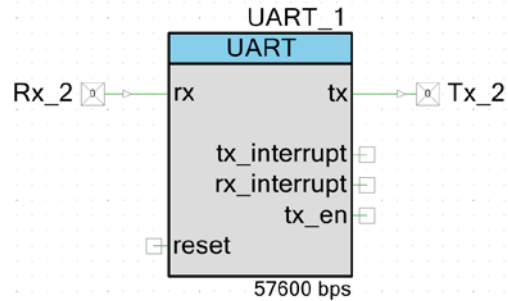
Hardware Connection between the KitProg and FreeSoC2 Target:

Debugger RX: Port [12]6 <-> Target Port[2]1 : UART TX

Debugger TX: Port[12]7 <-> Target Port[2]0 : UART RX

[Schematic Free SoC2](#)

How to read: *D0/1* ?



All these is set in the
component
Configuration Tabs

Features

- Baud rates from 110 to 921600 bps or arbitrary up to 4 Mbps
- RX and TX buffers = 4 to 65535
- Detection of Framing, Parity, and Overrun errors
- Full Duplex, Half Duplex, TX only, and RX only optimized hardware
- 8x or 16x oversampling

UART Configuration Tabs

Configure 'UART'

Name: **UART**

Configure | Advanced | Built-in

Mode

☒ Full UART (TX + RX) ☐ RX only

☐ Half duplex ☐ TX only

Bits per second:

Data bits:

Parity type:

☐ API control enabled

Stop bits:

Flow control:

API

1. UART_Start();
2. Reading / Writing blocking / non-blocking
3. Polling the status register / Interrupts for received data

Resource

[Component Datasheet "Universal Asynchronous Receiver Transmitter \(UART\)"](#)

Agenda

- o UART
- o Exercise

1. Exercise UART Talk Back

Implement

Develop a software on the PSoC, which receives the data from a terminal program on the windows and writes it back to the terminal in capital letters.

Step1: Write Back the text

Step2: Capatilize it

Use

- UART Component
- KitProg USB – UART Bridge
- Use Termite or putty as a terminal program.

Note

- Select maximum speed of 115200 Byte (limited by the KitProg)
- See the settings on KitProg User Guide

Termite Settings for “5. USB-UART” Bridge

Serial port settings

Port configuration

Port: COM5

Baud rate: 9600

Data bits: 8

Stop bits: 1

Parity: none

Flow control: none

Forward: none

Transmitted text

☐ Append nothing

☐ Append CR

☒ Append LF

☐ Append CR-LF

☒ Local echo

Options

☐ Stay on top

☒ Quit on Escape

☒ Autocomplete edit line

☒ Keep history

☐ Close port when inactive

Received text

Polling: 100 ms

Max. lines:

Font: default

☐ Word wrap

Plug-ins

☐ Auto Reply

☐ Function Keys

☐ Hex View

☐ Highlight

☐ Log File

User interface language: English (en)

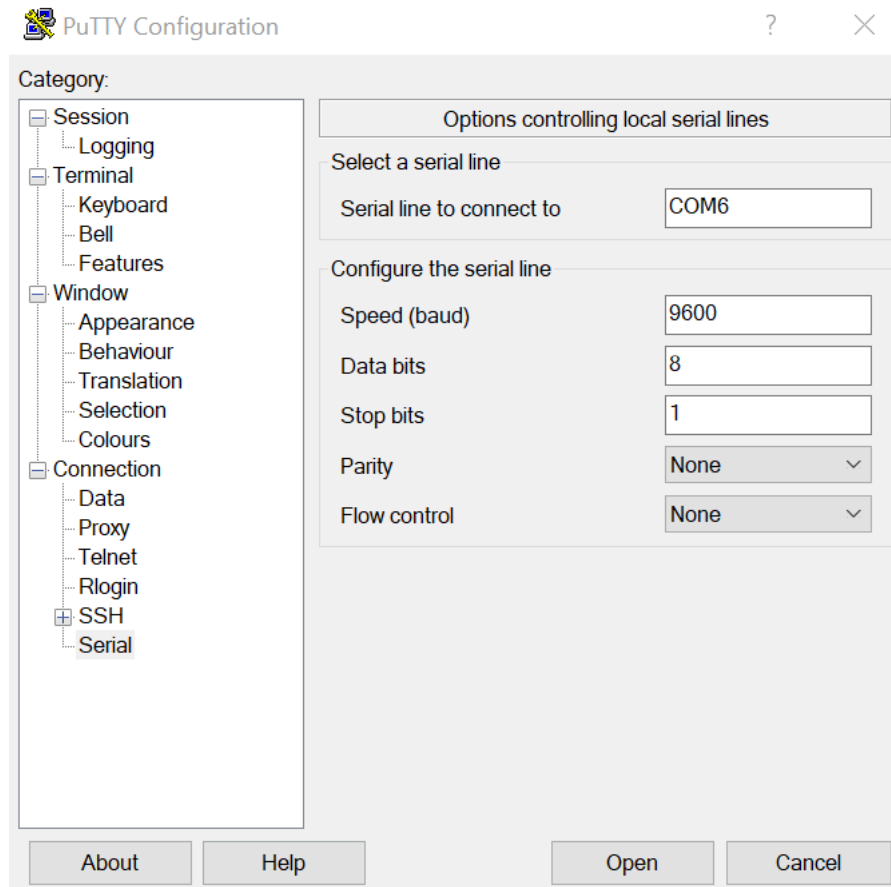
Cancel OK

[Step by Step: KitProg User Guide → 5. USB-UART Bridge](#)

Errata KitProg User Guide Rev *1

- UART Block for PSoC 5LP is vesion 2.5 and looks different, settings must be the same
- Typo in main.c in the pdf: UART_~~Uart~~PutString("CY8CKIT-044 USB-UART");
- Pin Assigment: see slide No. 5

Hints
for Your
Project



Additional Settings in Putty

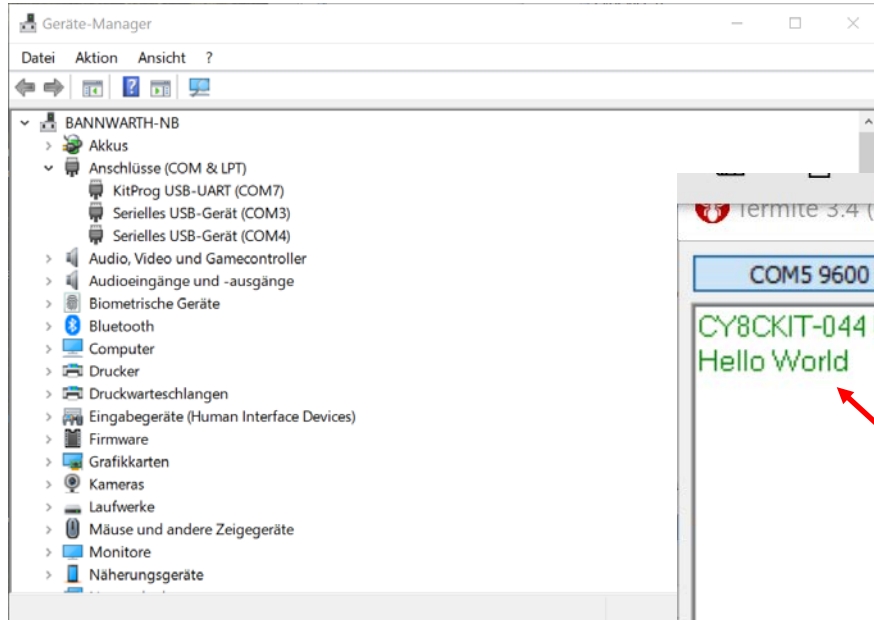
Terminal

Implicit LF in every LF
Local echo: force
Local line editing: force

Serial

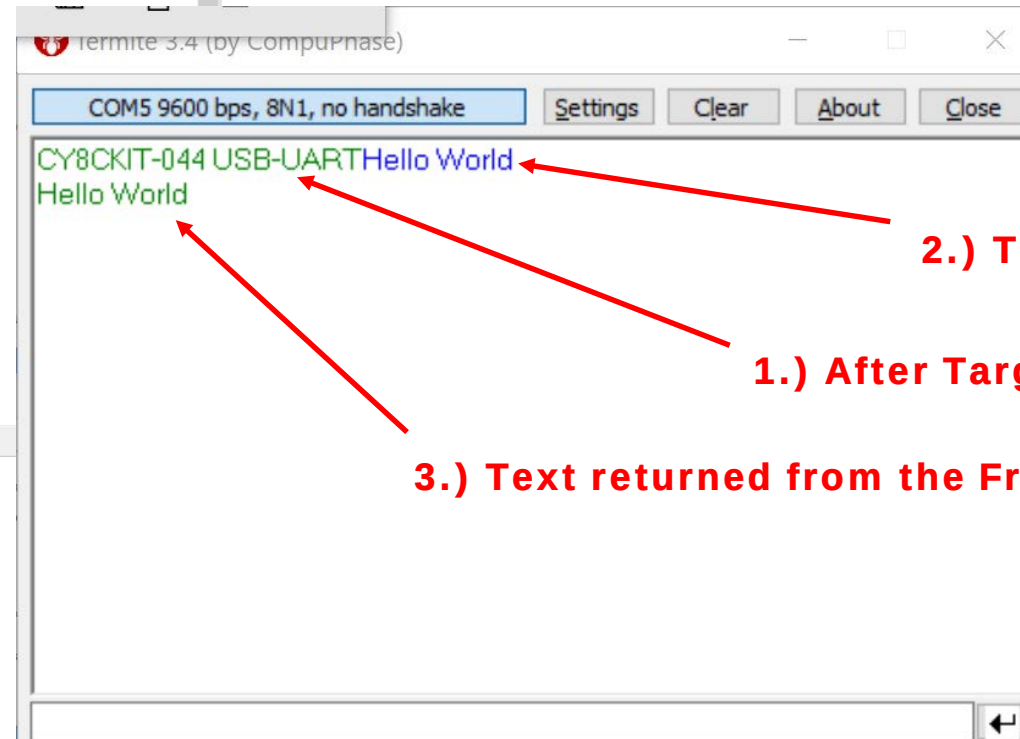
Flow control: none

Device Manager under Windows



→ get the COM Port Number

Termite Window

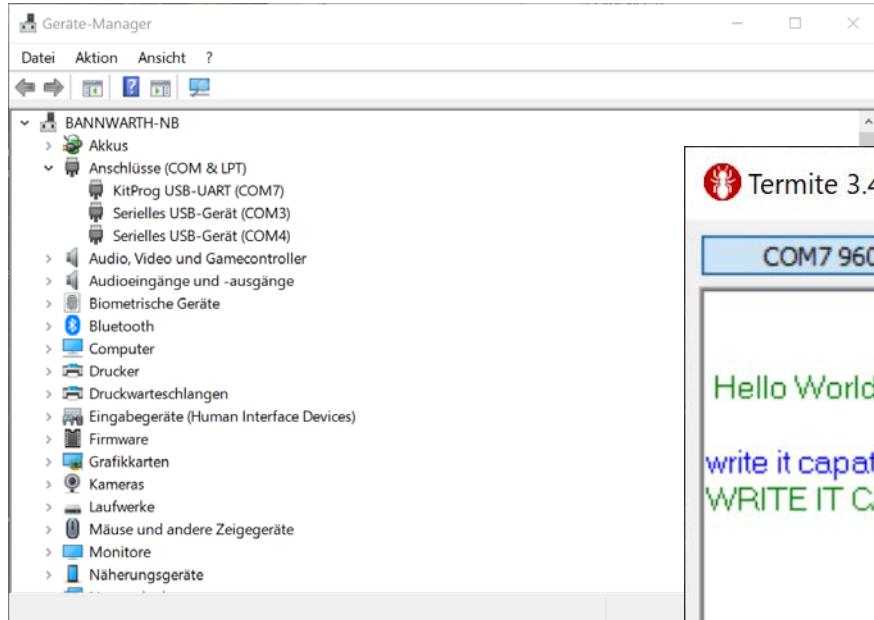


2.) Text entered in the input line

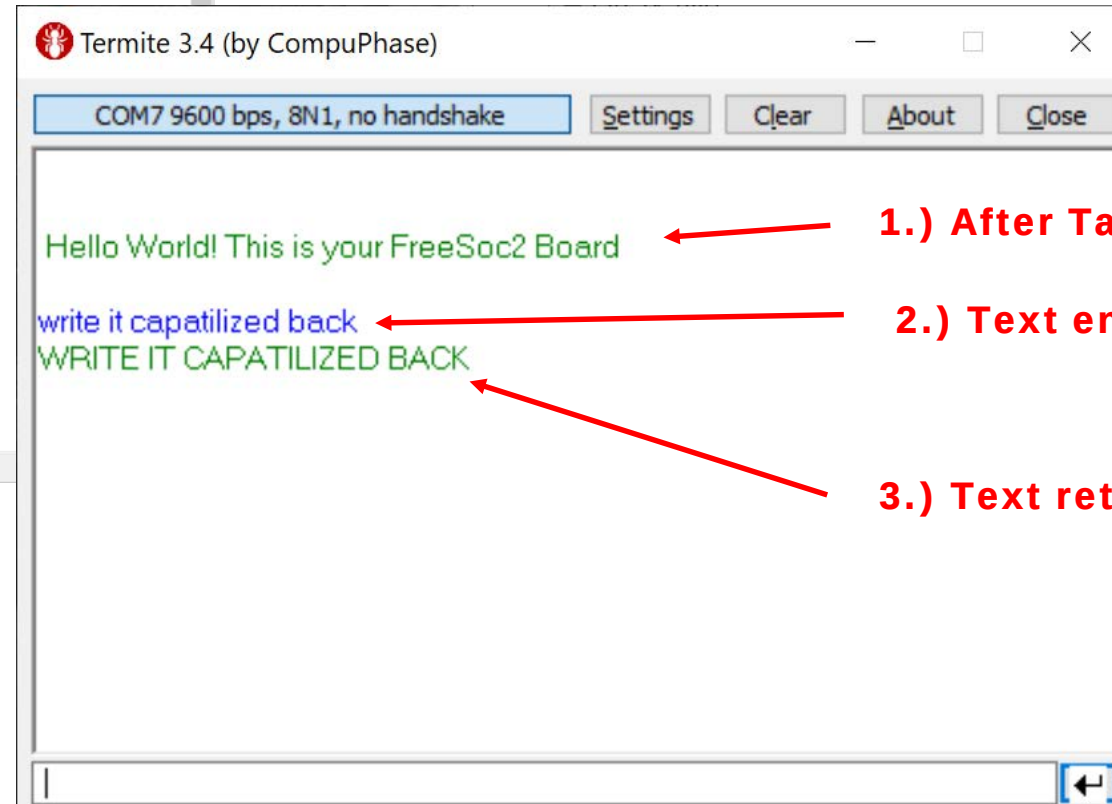
1.) After Target Reset

3.) Text returned from the FreeSoc 2

Device Manager under Windows



Termite Window



1.) After Target Reset

2.) Text entered in the input line

3.) Text returned from the FreeSoc 2

→ get the COM Port Number

→ See Moodle!

Literature

- “*Back to Basics: The Universal Asynchronous Receiver/Transmitter (UART)*”, by Robert Keim , [Link](#)
- “[Universal Asynchronous Receiver Transmitter \(UART\)](#)” Cypress, PSoC® Creator™ Component Datasheet
- “KitProg User Guide” [see section: 5. USB UART Bridge](#), Cypress, Userguide

Schematic:

- [Schematic Free SoC2](#)